Does Outside Money Win Elections?

Super PAC spending and electoral outcomes in 2024 U.S. House races

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In 2024, Super PAC money flowed overwhelmingly toward close races and candidates who were already strong, not toward long shots. Candidates with Super PAC backing won **55%** of competitive races versus **41%** without it. But once incumbency and party are taken into account, outside money becomes a *secondary* predictor of victory, behind both. The clearest reading of the data: **Super PAC spending marks winners more than it makes them.**

Background

Since the Supreme Court's *Citizens United* decision in 2010, Super PACs, independent expenditure-only committees that can raise and spend unlimited sums, have reshaped American elections. Outside spending on federal races reached a record **$4.5 billion** in 2024. Critics argue this gives wealthy donors outsized influence over who wins; defenders counter that the money simply chases candidates who are already viable. This report tests that question against the data.

The analysis is organized around four questions:

1. **Does Super PAC spending track electoral outcomes** — vote share and win/loss?
2. **Is it more important than a candidate's own fundraising?**
3. **Where does the money flow** — competitive or safe districts?
4. **Do Democrats and Republicans deploy it differently?**

A note on causation. Every relationship reported here is *associative, not causal*. Super PACs deliberately back candidates who are already competitive, so outside money and a candidate's underlying strength are entangled by design — no observational analysis can fully separate the two. Rather than overstate what the data can show, this report treats that limitation as part of the finding.

The data

The analysis combines three public datasets covering the 2024 U.S. House:

- **MIT Election Lab — U.S. House returns (1976–2024).** Certified, district-level general-election results from the Harvard Dataverse. We use the 2024 general election to compute each district's two-party vote share, margin of victory, and winner.

- **FEC Independent Expenditures (2023–24 cycle).** Transaction-level records of money spent *for* (type 24E) and *against* (type 24A) federal candidates by outside groups. We keep only committees registered as Independent Expenditure-Only (committee type "O") — i.e. Super PACs — and aggregate their spending to the candidate level.
- **FEC Candidate Summary (2024).** Each candidate's total disbursements (their own campaign spending) and incumbency status, used both as a control and to answer the fundraising question.

How competitiveness and spending are defined

District competitiveness is classified by the two-party margin of victory: **Toss-up** (0–5 points), **Competitive** (5–15), **Likely Safe** (15–30), and **Safe** (30+). This lets us ask not just *how much* outside money flows, but *where*, and whether it concentrates where races are genuinely in play.

Super PAC spending is the sum of independent expenditures by type-"O" committees on a given candidate, split into money spent *for* the candidate and money spent *against* them. Both directions matter: a candidate can be boosted by allies and attacked by opponents, and the two carry very different strategic logic.

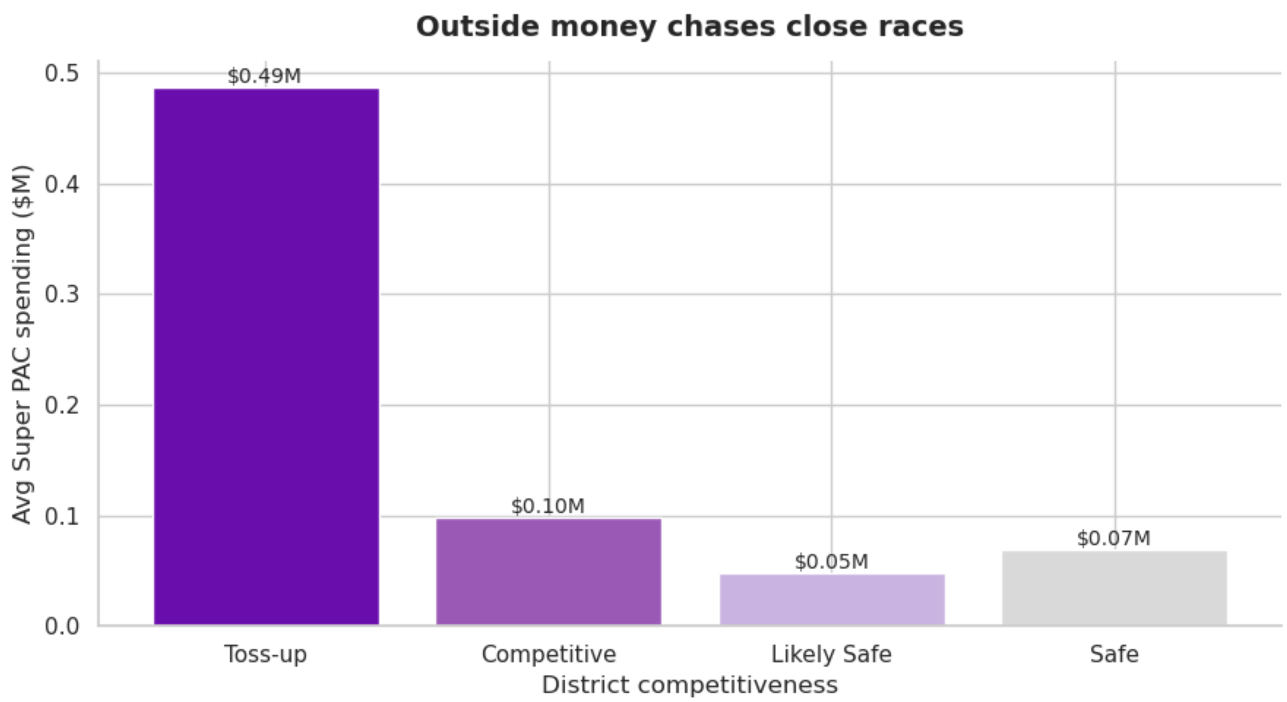
Building the analysis dataset

From the raw files we keep only 2024 general-election results, compute each district's vote share and margin, and classify competitiveness. On the finance side, we isolate independent expenditures (24E/24A) made by Super PACs and aggregate them per candidate into spending *for* and *against*. We then merge in each candidate's own disbursements and incumbency status. The result is one row per major-party candidate combining electoral outcome, outside spending, and self-funding, the single table behind every query and chart below. It is also written to a SQLite database so the relationships can be explored with SQL.

1. Where does the money go?

If outside money were spread evenly, it would tell us little about strategy. It is not. The queries below trace Super PAC spending across district competitiveness, the highest-spending districts, and how concentrated the money is within each state.

	competitiveness	num_districts	avg_superpac	total_superpac
0	Toss-up	38	486711.94	162561788.0
1	Competitive	71	97913.38	59433424.0
2	Safe	205	69639.14	78761869.0
3	Likely Safe	122	48334.86	37894527.0



	CAND_OFFICE_ST	CAND_OFFICE_DISTRICT	competitiveness	winner_party	total_su
0	NY	16.0	Safe	DEMOCRAT	1683
1	AK	0.0	Toss-up	REPUBLICAN	1576
2	CA	47.0	Toss-up	DEMOCRAT	1414
3	NY	19.0	Toss-up	DEMOCRAT	116(
4	MO	1.0	Safe	DEMOCRAT	116(
5	CA	45.0	Toss-up	DEMOCRAT	873
6	CO	8.0	Toss-up	REPUBLICAN	828
7	SC	1.0	Likely Safe	REPUBLICAN	738
8	NE	2.0	Toss-up	REPUBLICAN	699
9	VA	7.0	Toss-up	DEMOCRAT	692

	group_label	num_districts	num_competitive	pct_competitive
0	Above average spending	130	73	56.2
1	Below average spending	306	36	11.8

	CAND_OFFICE_ST	CAND_OFFICE_DISTRICT	competitiveness	district_superpac	pct
0	AK	0.0	Toss-up	15764631.0	
1	AL	2.0	Competitive	3124498.0	
2	AL	1.0	Safe	2731059.0	
3	AL	7.0	Likely Safe	75926.0	
4	AL	3.0	Safe	0.0	
5	AL	4.0	Safe	0.0	
6	AL	5.0	Safe	0.0	
7	AL	6.0	Safe	0.0	
8	AR	3.0	Safe	188992.0	
9	AR	2.0	Likely Safe	156346.0	
10	AR	4.0	Safe	1282.0	
11	AR	1.0	Safe	0.0	
12	AZ	6.0	Toss-up	5444398.0	
13	AZ	1.0	Toss-up	4278232.0	
14	AZ	3.0	Safe	2030345.0	
15	AZ	8.0	Competitive	1857783.0	
16	AZ	2.0	Competitive	249473.0	
17	AZ	4.0	Competitive	41634.0	
18	AZ	7.0	Likely Safe	682.0	
19	AZ	5.0	Likely Safe	0.0	

Super PACs spend where the race is close. Toss-up districts drew an average of about **$486K** in Super PAC spending per candidate; Safe districts drew **$69K** — roughly seven times less, and the drop-off is smooth across all four categories. The pattern holds when we flip the question around: among districts with above-average outside spending, **56%** were genuinely competitive, versus just **12%** of low-spending districts, a five-fold difference.

The money is concentrated even *within* states. In most states, one or two districts absorb nearly all of the outside dollars while safe seats receive little or nothing (in Alabama, two districts split 99% of the state's total). New York's 16th led the entire country at **$16.8M**, driven by a high-profile primary that pulled in national money. The picture is of Super PACs behaving like rational investors: **they put their money where the outcome is genuinely in doubt**.

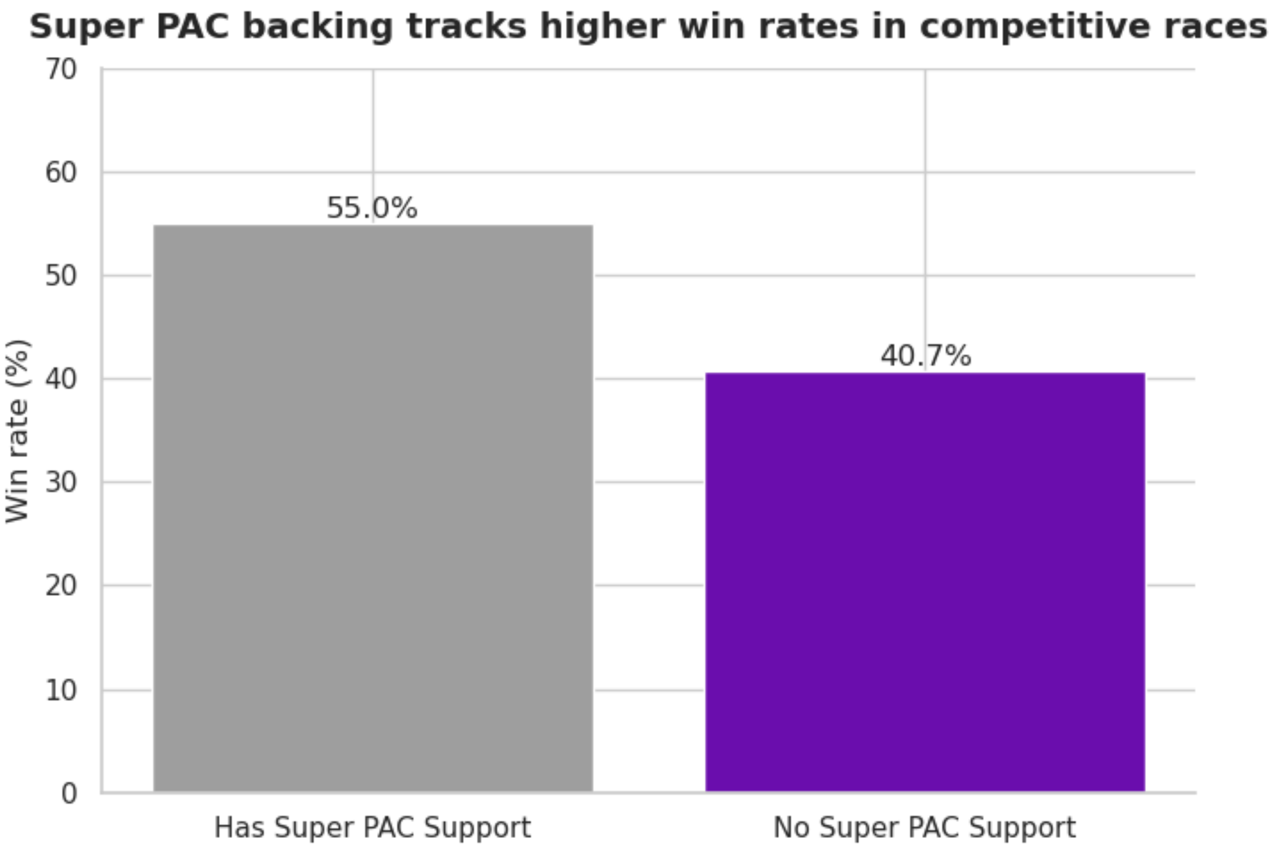
2. Does the money win?

If outside money chases close races, the natural next question is whether it tips them. We begin with the simplest possible comparison, then control for the two factors that obviously confound it — **incumbency and party**.

Group	Candidates	Wins	Win rate (%)
Has Super PAC Support	229	126	55.0%
No Super PAC Support	595	242	40.7%

The raw gap is real: in competitive districts, candidates with any Super PAC support in their favor won **55%** of the time, versus **40.7%** for those with none — a 14-point spread that matters when margins are often in single digits.

But a raw gap cannot tell us whether the money *won* those races or simply *picked* candidates who were already likely to win. To separate the two, we model outcomes while holding incumbency and party constant.



	CAND_PTY_AFFILIATION	competitiveness	outcome	num_candidates	avg_superpac
0	DEM	Competitive	Loser	102	3265
1	DEM	Likely Safe	Loser	204	103
2	DEM	Safe	Loser	197	3
3	DEM	Toss-up	Loser	76	29625
4	DEM	Competitive	Winner	162	13299
5	DEM	Likely Safe	Winner	142	5294
6	DEM	Safe	Winner	340	6486
7	DEM	Toss-up	Winner	83	35812
8	REP	Competitive	Loser	235	1285
9	REP	Likely Safe	Loser	146	75
10	REP	Safe	Loser	208	58
11	REP	Toss-up	Loser	111	17154
12	REP	Competitive	Winner	108	13041
13	REP	Likely Safe	Winner	292	5255
14	REP	Safe	Winner	386	3945
15	REP	Toss-up	Winner	64	23444

Winners consistently drew more outside support than losers, Democratic winners averaged about **$111K** versus **$45K** for losing Democrats; Republican winners **$70K** versus **$31K**. But notice that the gap in candidates' *own* spending was even larger. That is the first hint that fundraising capacity, not outside money, is doing much of the work.

To pin this down, we fit two models on competitive districts only — **Toss-ups and Competitive** seats, where money is most likely to matter (Safe districts are excluded because their outcomes are driven by partisan lean, not spending). Each model uses six predictors: Super PAC spending for and against the candidate, the candidate's own spending, an indicator for any Super PAC support, party, and incumbency. Spending variables are log-transformed because campaign finance is heavily right-skewed.

```
=== Model 1: Linear Regression (predicting margin of victory) ===
R²: 0.143   |   RMSE: 8.007
```

Predictor	Coefficient
is_incumbent	2.858
log_superpac_for	2.087
party_dem	1.395
log_TTL_DISB	0.150
log_superpac_against	-0.053
has_superpac	-2.126

Model 1 — Linear regression (margin of victory). The model explains about **14%** of the variation in margin ($R^2 = 0.143$; average error roughly 8 points), modest, as expected for something as complex as an election, but enough to rank the predictors. **Incumbency is the strongest** (+2.86 points), followed by party (+1.40 for Democrats in 2024's competitive seats). Super PAC spending *for* a candidate lands between them (+2.09) — a real, positive association. The candidate's own spending adds little once everything else is held constant (+0.15), and spending *against* a candidate has essentially no effect (–0.05).

```
=== Model 2: Logistic Regression (predicting win/loss) ===
Accuracy: 0.667
```

	precision	recall	f1-score	support
0	0.74	0.62	0.67	91.00
1	0.61	0.73	0.66	74.00
accuracy	0.67	0.67	0.67	0.67
macro avg	0.67	0.67	0.67	165.00
weighted avg	0.68	0.67	0.67	165.00

Predictor	Coefficient
is_incumbent	0.828
party_dem	0.548
log_superpac_for	0.534
log_superpac_against	0.075
log_TTL_DISB	-0.052
has_superpac	-0.599

Model 2 — Logistic regression (win/loss). The model picks the winner in **67%** of races, well above a 50% coin flip, with precision and recall balanced across both classes (so it is not simply guessing the larger group). The ordering mirrors Model 1: **incumbency first** (0.83), then party (0.55), then Super PAC support (0.53), with own spending near zero (–0.05) and, notably, spending *against* a candidate carrying a small *positive* coefficient (+0.075). Being attacked by outside money does not predict losing; if anything, it signals a candidate worth attacking.

Both models tell one story, and the chart makes it visual: **structure beats money.** Who you are (an incumbent) and which party you run under in a given cycle explain more about winning than how much outside money shows up. Super PAC support has a real, positive, but secondary pull.

The scatter drives the point home. At every level of Super PAC spending, candidates land on both sides of the win/loss line — some win comfortably, others lose badly with similar backing. The wide dispersion is exactly what the low R^2 predicted: outside money alone is a weak guide to *how much* a candidate wins or loses by. The Republican trend line is steeper but starts lower (many GOP candidates in competitive seats received little support and lost), while the Democratic line is flatter and higher, reflecting Democrats' stronger baseline in 2024's competitive races.

3. Does outside money replace a candidate's own fundraising?

A common assumption is that Super PACs let candidates outsource their fundraising, that outside money substitutes for a strong campaign. The data say the opposite.

	superpac_support_level	num_candidates	avg_candidate_spending	avg_superpac	c
0	High Super PAC support	137	4507559.34	2245989.85	
1	Low Super PAC support	488	2263514.80	63509.17	
2	No Super PAC support	2231	228501.30	0.00	

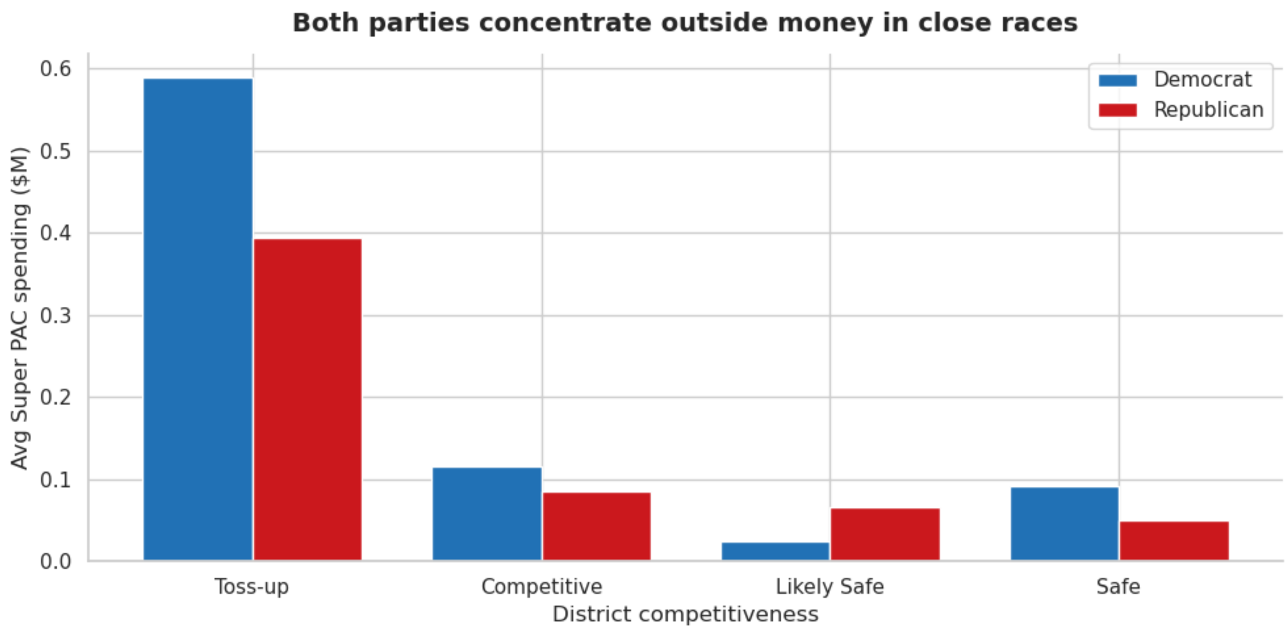
	CAND_PTY_AFFILIATION	competitiveness	num_districts	avg_superpac	avg_candida
0	DEM	Toss-up	38	588755.02	
1	DEM	Competitive	71	115381.77	
2	DEM	Safe	196	91988.57	
3	DEM	Likely Safe	120	25056.91	
4	REP	Toss-up	38	393998.51	
5	REP	Competitive	71	84468.33	
6	REP	Likely Safe	122	66723.37	
7	REP	Safe	191	49434.35	

Outside money *complements* a strong campaign; it does not replace one. Candidates with high Super PAC support spent about **$4.5M** of their own money on average, versus **$2.3M** for those with low support and just **$228K** for those with none. The money piles onto strength rather than standing in for it, and in the models above, a candidate's own disbursements out-predicted Super PAC support once structural factors were controlled for. The answer to Question 2 is clean: **outside money is not more important than a candidate's own fundraising.**

4. Democrats vs. Republicans

Both parties target the same districts; they differ mainly in how much they rely on outside money.

	CAND_PTY_AFFILIATION	num_candidates	total_superpac	avg_superpac
0	DEM	1328	182140390.0	137153.91
1	REP	1569	161250579.0	102772.84



Democrats pulled in more Super PAC dollars in absolute terms in competitive districts. But Republicans were *relatively* more dependent on outside money, it made up a larger share of their total campaign resources (about **35%** of own spending in toss-ups, versus **25%** for Democrats). Both parties concentrate outside money in the same place: toss-up and competitive seats. The disagreement is over how much to lean on it, not where to aim it.

Conclusion

Across every angle, the 2024 House data point the same way: **Super PAC money follows competitiveness and candidate strength rather than creating winners.**

- **Does it win?** On the surface, yes, 55% vs 41% in competitive districts. But once incumbency and party are controlled for, outside money is only a secondary predictor. Super PACs back candidates who are already positioned to win, which makes it hard to credit the money with the victory.
- **More important than own fundraising?** No. A candidate's own spending out-predicted Super PAC support, and outside money tended to pile onto already well-funded campaigns.
- **Where does it go?** Where races are close. Toss-up districts drew roughly five times the outside money of safe ones, and within most states one or two districts absorbed nearly all of it.
- **Partisan differences?** More style than scale. Democrats raised more outside money outright; Republicans leaned on it more heavily as a share of resources. Both concentrated it in competitive seats.

Taken together, this is the **endogeneity problem** that runs through campaign-finance research: money flows to viable candidates, so its independent effect is genuinely difficult to isolate, and honest analysis should say so rather than overclaim.

Next steps. Add Cook Political Report PVI scores to control directly for district partisan lean, the key omitted variable here. Extend the analysis to the 2022 cycle to test whether the patterns hold across electoral environments. And examine primaries, where Super PAC spending may move less predictable outcomes than it does in general elections.

Appendix

Supporting checks and additional queries behind the analysis above: data-quality validation, the spending distribution, and full summary tables.

Data quality

The modeling dataset has no major missing-value problems — only `CAND_ICI` (incumbency) carried a handful of missing values, and every variable used in the models is complete. There are no problematic duplicate rows; numeric finance variables are stored as numbers, and the binary `won` and `is_incumbent` indicators are correctly encoded.

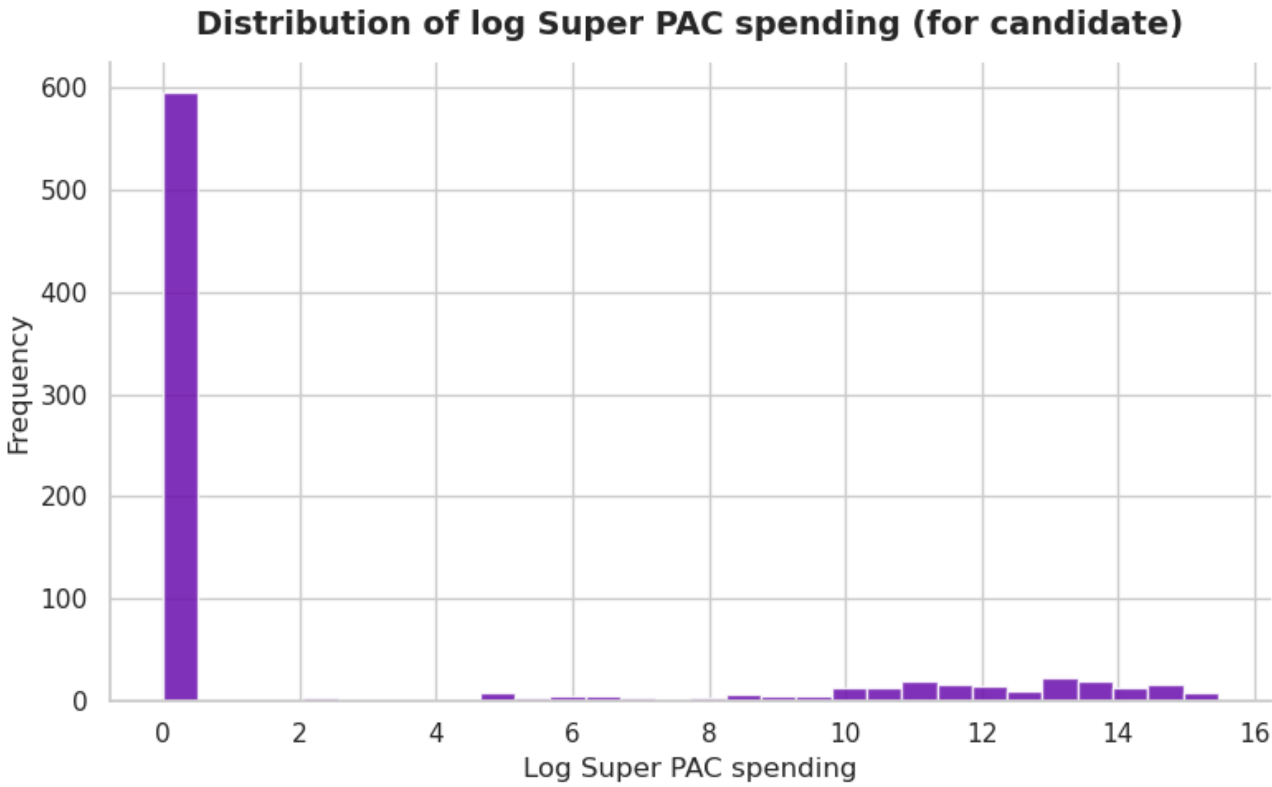
```
CAND_ID                 0
CAND_NAME               0
CAND_PTY_AFFILIATION    0
CAND_OFFICE_ST          0
CAND_OFFICE_DISTRICT    0
TTL_DISB                0
superpac_for            0
superpac_against        0
superpac_total          0
dem_pct                 0
rep_pct                 0
margin                  0
competitiveness         0
winner_party            0
CAND_ICI                5
is_incumbent            0
margin_outcome          0
won                     0
has_superpac            0
log_superpac_for        0
log_superpac_against    0
log_TTL_DISB            0
party_dem               0
has_superpac_label      0
dtype: int64
```

```
np.int64(0)
```

```
log_superpac_for        float64
log_superpac_against    float64
log_TTL_DISB            float64
won                       int64
is_incumbent              int64
dtype: object
```

Spending distribution and outliers

Super PAC spending is heavily right-skewed, which is why we log-transform it before modeling. The extreme values are real, a handful of districts genuinely attract enormous sums, so they are kept rather than trimmed; they represent meaningful variation, not data errors.



Summary tables

	competitiveness	count	mean	median	min	max
0	Competitive	607	97913.38	0.0	-14629.0	5765534.0
1	Likely Safe	784	48334.86	0.0	-1357.0	4537946.0
2	Safe	1131	69639.14	0.0	-21841.0	12110317.0
3	Toss-up	334	486711.94	0.0	0.0	11510742.0

	CAND_PTY_AFFILIATION	num_candidates	avg_superpac	median_superpac	avg_car
0	DEM	1306	139464.31	0.0	
1	REP	1550	100974.98	0.0	